

EDAN20 - Assignment 0

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1 Spelling Corrector

1.1 First Impression

I looked mostly on the original spell corrector for this assignment, the first thing that I thought about was how simple and elegant the solution was. When i read the introduction I also though about how I would build the program to solve the same problems. One thing I had problems with was the equivalent solution to *edits1*, which is a nice was of solving it.

1.2 Tokenization

Another thing I noted about the program was how dependent it is on the tokenization, *re.findall(r'\w+', text.lower())* is a simple way to tokenize which treats sequences of letters, digits and underscores as a *word*. This works for English, but it also ignores punctuation and hyphens which means that the model's understanding of the input text is limited right from the start.

1.3 The Corpus

Another thing with this spell corrector is that the dictionary *WORDS* is built from the file *big.txt*. The probability of a word is just the frequency in that corpus. The model inherits all the biases of the corpus and since it has not been updated for a while it is not that reliable for modern slang, technical terminology or specialized vocabulary, but still useful for typical English words.

1.4 P(word)

The probability function used in the program $P(word)$ assigns likelihoods to words without considering context or syntax. This is very simple compared to modern spell correctors. But it still captured a important insight, that some words are far more common than others and this frequency can guide the correction. Since it does not care about the context it cannot make the correct decision in a real world problem such as "*form*" vs "*from*", which requires understanding of the surrounding sentence.

1.5

As I mentioned earlier the program has a great strength in the *edit* functions that can handle common human spelling errors. The functions generate possible corrections based on typical typing mistakes such as deletions, insertions, replacements and transpositions. This reflects real human behavior. But as also mentioned above it can not correct real world errors because both words exists in the dictionary.